

Prophet Inequalities Over Time

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1 Introduction

The main sections that this report is going to have are the following :

1. The problem statement
2. Why $n \rightarrow \infty$ is a hard instance.
3. The naive algorithm
4. Approach 1 that is taking it to continuous domain, telling the optimal algorithm and leaving at the optimal payoff that can be maximised using functional extremisation
5. Approach 2 that is constructing the worst case distribution
6. Establishing the new form of ASDness incorporating LP Duality in its proof.
7. Analysing the quantile algorithm by the authors and why it works
8. After establishing the ASD guarantee we can also say that the best case choice of p looks like what, tell its conditions and its differential equation.

2 Problem statement

Suppose you have to rent a room for n days. On each of the n days, a buyer comes, offers you a price that he/she is willing to pay per day. You need to decide the number of days that you want to rent the room to the buyer. The buyer will take the room for as many days as you offer him. You cannot rent the room to buyer i if the room has already been rented on day i to some previous buyer. The distribution of the prices offered by the buyer is known. We need to construct the optimal strategy to rent the room and compare it against an adversary that already knows the values that the buyers are going to offer.

Specifically, we are dealing with the scenario where the buyers are *iid* (independent and identically distributed).

2.1 Formal statement

Model the value of the i^{th} person as a random variable X_i . X_i 's are assumed to be iid variables with the distribution F . The items will be observed in the order : $x_n, x_{n-1}, \dots, x_1$. For each x_i decide the number of days d_i to choose it. The value obtained by the algorithm is thus given by :

$$ALG = \sum_{i=n}^1 (x_i \cdot d_i) \quad (1)$$

Since we are not allowed to rent a room on day for which it has already been rented, we have the constraint that if $d_i = k$, then $d_{i-1}, d_{i-2}, \dots, d_{i-k} = 0$.

Let the payoff obtained by the prophet be denoted by OPT . Then the competitive ratio is defined as follows :

$$\text{Competitive Ratio} = \frac{\mathbb{E}[ALG]}{\mathbb{E}[OPT]} \quad (2)$$

The expectation is over the random variables $X_1, X_2, \dots, X_n$ and over the algorithm (if it is randomised). Our goal is to come up with an algorithm that maximises the competitive ratio, over the worst case n and F .

Also note that we only deal with distributions with positive support, therefore $F(x) = 0 \forall x < 0$. This also allows us to use 22 to express the expected value of the random variables.

3 Optimal strategy for the prophet

Since the prophet knows the values in advance, he rents a room to a buyer till the day when a buyer with a better values arrives. This means on day i , the rent earned by the prophet is $\max(x_n, x_{n-1}, \dots, x_i)$ The payoff obtained by the prophet can be expressed as follows :

$$OPT = \sum_{i=n}^1 \max(x_n, x_{n-1}, \dots, x_i) \quad (3)$$

$$\mathbb{E}[OPT] = \sum_{i=1}^n \mathbb{E}[\max(x_n, x_{n-1}, \dots, x_i)] \quad (4)$$

Since the max of n random variables with distributions F is distributed according to F^n

$$\begin{aligned} \mathbb{E}[OPT] &= \sum_{i=1}^n \int_{x=0}^{\infty} (1 - F^i(x)) dx \\ &= \int_{x=0}^{\infty} \left(n - \sum_{i=1}^n (F^i(x)) \right) dx \\ &= \int_{x=0}^{\infty} \left(n - F(x) \cdot \frac{(1 - F^n(x))}{1 - F(x)} \right) dx \end{aligned} \quad (5)$$

4 Previous Work

This problem was first introduced in this paper by Abels Et al. [1]. They gave the following important results :

1. The optimal dynamic programming algorithm
2. An upper bound on the competitive ratio of $1/\phi \approx 0.618$
3. A quantile-based algorithm, along with a lower bound of 0.598 on the CR of this algorithm

1 and 3 were in particular interest to us.

The dynamic programming algorithm uses distinct threshold for each day defined as follows : $\tau_0 = 0$, and $\tau_i = \frac{\mathbb{E}[\text{Payoff of algorithm from } i \text{ to } 1]}{i}$. Note that the thresholds represent the average daily payoff of the algorithm from that time step. The online decision process works as follows : Abels Et al.

Algorithm 1: Optimal Online Decision Procedure

```

1 for  $i = n, \dots, 1$  do
2   if  $x_i \geq \tau_{i-1}$  then
3      $\lfloor$  Select  $x_i$  for all remaining  $i$  time steps ;
4   else
5      $\lfloor$  Select  $x_i$  only in step  $i$ 

```

proved that the τ_i 's defined in the above fashion form an increasing sequence, and hence established the optimality of the algorithm. See appendix B. They, however, were unable to derive a lower bound for this algorithm, nor were they able to obtain a closed form of the threshold to work with.

5 Determining if $n \rightarrow \infty$ is the worst case

In the single choice iid prophet problem it is observed that $n \rightarrow \infty$ is the worst case, where we simulate an algorithm on n items using an algorithm on $2n$ items and observing that the competitive ratio improves on n as compared to $2n$.

Suppose we have an Algorithm that guarantees CR_{2n} on $2n$ items **complete this**

Hence we establish that, $\alpha_n \geq \alpha^*(1 - \frac{1}{n})$ Although this does not imply that $n \rightarrow \infty$ is the worst case, it does motivate us to consider the case because this tells us that worst case n needs to be a large value.

6 Analysing $n \rightarrow \infty$

Consider the thresholds described in 3. Let the distribution of the maximum of the random variables be F . Therefore the distribution of each random variable is $F^{1/n}$. The value of the thresholds can

70 be expressed as follows :

$$\begin{aligned}
\tau(i+1) &= \frac{1}{i+1} \cdot [\mathbb{P}[X < \tau(i)] \cdot (\mathbb{E}[X|X < \tau(i)] + i\tau(i)) + \mathbb{P}[X \geq \tau(i)] \cdot ((i+1)\mathbb{E}[X|X \geq \tau(i)])] \\
&= \frac{1}{i+1} \cdot [\mathbb{P}[X < \tau(i)] \cdot (\mathbb{E}[X|X < \tau(i)]) + \mathbb{P}[X \geq \tau(i)] \cdot (\mathbb{E}[X|X \geq \tau(i)]) \\
&\quad + i \cdot (\mathbb{P}[X < \tau(i)] \cdot \tau(i) + \mathbb{P}[X \geq \tau(i)] \cdot \mathbb{E}[X|X > \tau(i)])] \\
&= \frac{1}{i+1} \cdot [\mathbb{E}[X] + i \cdot (\mathbb{P}[X < \tau(i)] \cdot \tau(i) + \mathbb{P}[X \geq \tau(i)] \cdot \mathbb{E}[X|X > \tau(i)])]
\end{aligned}$$

Keep in mind that the $\mathbb{E}[X]$ term above comes from picking the item on day i , it counts the contribution of a) picking x_i when we don't commit and b) contribution of x_i on day i when we
71 commit

$$= \frac{1}{i+1} \cdot \left[\mathbb{E}[X] + i \cdot \left(\tau(i) + \int_{x=\tau(i)}^{\infty} (1 - F^{1/n}(x)) dx \right) \right] \quad (6)$$

72 Multiplying by $(i+1)$ and subtracting $i \cdot \tau(i)$, we get

$$(i+1) \cdot \tau(i+1) - (i) \cdot \tau(i) = \mathbb{E}[X] + i \cdot \left(\int_{x=\tau(i)}^{\infty} (1 - F^{1/n}(x)) dx \right)$$

73 For a large n , let $t = i/n$, substituting above we get :

$$(nt+1) \cdot \tau(nt+1) - (nt) \cdot \tau(nt) = \mathbb{E}[X] + nt \cdot \left(\int_{x=\tau(nt)}^{\infty} (1 - F^{1/n}(x)) dx \right)$$

74 dividing by n on both sides, we get :

$$\left(t + \frac{1}{n}\right) \cdot \tau\left(n\left(t + \frac{1}{n}\right)\right) - t \cdot \tau(nt) = \frac{\mathbb{E}[X]}{n} + t \cdot \left(\int_{x=\tau(nt)}^{\infty} (1 - F^{1/n}(x)) dx \right)$$

75 Also note that, as $n \rightarrow \infty$:

$$1 - F^{1/n}(x) = \frac{1}{n} \cdot \ln \left(\frac{1}{F(x)} \right)$$

76 Let $1/n = dt$

$$(t + dt) \cdot \tau(n(t + dt)) - t \cdot \tau(nt) = \mathbb{E}[X]dt + t \cdot \left(\int_{x=\tau(nt)}^{\infty} \ln \left(\frac{1}{F(x)} \right) dt dx \right)$$

$$\frac{(t + dt) \cdot \tau(n(t + dt)) - t \cdot \tau(nt)}{dt} = \mathbb{E}[X] + t \cdot \left(\int_{x=\tau(nt)}^{\infty} \ln \left(\frac{1}{F(x)} \right) dx \right)$$

77 We overload notation and let $\tau(t)$ denote $\tau(\lfloor nt \rfloor)$ for $0 \leq t \leq 1$

$$\frac{(t + dt) \cdot \tau(t + dt) - t \cdot \tau(t)}{dt} = \mathbb{E}[X] + t \cdot \left(\int_{x=\tau(t)}^{\infty} \ln \left(\frac{1}{F(x)} \right) dx \right)$$

$$\frac{d}{dt}(t \cdot \tau(t)) = \mathbb{E}[X] + t \cdot \left(\int_{x=\tau(t)}^{\infty} \ln \left(\frac{1}{F(x)} \right) dx \right)$$

78 Note that the expected value of X given by

$$\mathbb{E}[X] = \int_{x=0}^{\infty} (1 - F^{1/n}(x)) dx$$

79 vanishes when $n \rightarrow \infty$.

80 **As discussed in eq. (6), this means that the contribution of picking the item for a single day is negligible when $n \rightarrow \infty$**

81 Finally, we obtain :

$$t \cdot \tau'(t) + \tau(t) = t \cdot \left(\int_{x=\tau(t)}^{\infty} \ln \left(\frac{1}{F(x)} \right) dx \right). \quad (7)$$

82 We can further differentiate, the above equation with respect to t , in order to get rid of the integral
83 to obtain :

$$F(\tau(t)) = \exp \left(\frac{t^2 \cdot \tau''(t) + t \cdot \tau'(t) - \tau(t)}{t^2 \cdot \tau'(t)} \right) \quad (8)$$

84 We can use 5 to express the payoff of the adversary, note that we need to $F^{1/n}$ instead of F in 5
 85 since the distribution of each item is given by $F^{1/n}$.

$$\begin{aligned}\mathbb{E}[OPT] &= \int_{x=0}^{\infty} \left(n - F^{1/n}(x) \cdot \frac{(1 - F(x))}{1 - F^{1/n}(x)} \right) dx \\ &= \int_{x=0}^{\infty} \left(n - \frac{(1 - F(x))}{1/n \cdot \ln(\frac{1}{F(x)})} \right) dx \\ &= n \cdot \int_{x=0}^{\infty} \left(1 - \frac{(1 - F(x))}{\ln(\frac{1}{F(x)})} \right) dx\end{aligned}$$

86 The competitive ratio can be expressed as follows :

$$\begin{aligned}\alpha &= \frac{\mathbb{E}[ALG]}{\mathbb{E}[OPT]} = \frac{n \cdot \tau(1)}{n \cdot \int_{x=0}^{\infty} \left(1 - \frac{(1 - F(x))}{\ln(\frac{1}{F(x)})} \right) dx} \\ \frac{1}{\alpha} &= \frac{\int_{x=0}^{\infty} \left(1 - \frac{(1 - F(x))}{\ln(\frac{1}{F(x)})} \right) dx}{\tau(1)}\end{aligned}$$

87 substituting x by $\tau(t)$ in the numerator we obtain :

$$\frac{1}{\alpha} = \frac{\int_{t=0}^{\infty} \left(1 - \frac{(1 - F(\tau(t)))}{\ln(\frac{1}{F(\tau(t)))} \right) \tau'(t) dt}{\tau(1)} \quad (9)$$

88 Using 8, we can express $1/\alpha$ as a function of $\tau(t)$ and t alone.

89 Using 7 we know that an F determines a unique $\tau(t)$, and a valid $\tau(t)$ corresponds to a unique CDF
 90 F . Therefore in order to find the minimum α , we can maximise the RHS over the space of valid
 91 $\tau(t)$'s. This is functional extremisation and methods such as calculus of variation can be applied.
 92 We will obtain the worst case $\tau(t)$ corresponding to the worst case F and its competitive ratio α .
 93 I attempted that but the expressions turned out to be pretty complicated.

94 7 Trying to construct the worst case F

95 8 Review of Abels Et al.'s algorithm

96 In their paper Abels Et al. present the following algorithm (Note that the change of observation
 97 order from $i = n, \dots, 1$ to $i = 1, \dots, n$.

98 The algorithm's threshold in step $i = 1, \dots, n$ is the quantile $\delta_{p(i)} = F^{-1}(p(i))$, where $p(i)$ denotes
 99 some probability. We let $p(0) = 1, p(n+1) = p(n+2) = 0$. They further prove that the algorithm
 100 ONL with the threshold $\delta_{p(i)}$ where $p(i) = \frac{-ci}{n^2}$ and $c = 9.71$ gives a lower bound of 0.598 for $n \rightarrow \infty$

Algorithm 2: ONL

```
1 for  $i = 1, \dots, n$  do
2   if  $x_i \geq \delta_{p(i)}$  then
3     Select  $x_i$  for all  $n-i+1$  remaining steps ;
4   else
5     Select  $x_i$  random variable for one step
```

9 Approximate Stochastic Dominance is equivalent to Competitiveness

Suppose we have an algorithm that guarantees the following:

$$\mathbb{E}[\text{time ALG chooses} \geq x] \geq c \cdot \mathbb{E}[\text{time OPT chooses} \geq x] \quad \forall x \geq 0 \quad (10)$$

We will call that this algorithm is a c -ASD algorithm. Note that the term ASD is usually used to compare distributions and has a different (but similar) definition. Whenever I use the term ASD in this text, it would mean as what is defined in 10. We will prove the following :

1. A c -ASD algorithm is c - competitive
2. If there exists a c competitive algorithm then there also exists a c -ASD algorithm.

Look at the appendix C.1 and ?? for a proof of the above statements.

Prove that this also extends to the non finite support cases, with arbitrary precision.

9.1 Benefit from ASDness

Once we have established that a c -ASD algorithm leads to a c -competitive algorithm and vice versa. This simplifies our process for constructing an algorithm. In particular we do not need to look for a worst case distribution. Instead we can work with any distribution of our choice, if we obtain an algorithm ALG that is c -competitive on F , then we can construct an algorithm ALG' for any other distribution F' , such that the thresholds used for decision making by ALG' satisfy the following property : If for the i -th item, the threshold used by ALG is τ , then ALG' uses the threshold τ' such that $F'(\tau') = F(\tau)$. **Explain the border cases such as when there are flat regions or jumps, also talk about randomisation**

10 Quantile based algorithm

For $n \rightarrow \infty$ we construct an algorithm for any general distribution F , that takes decisions based on the quantile of the distribution the element lies in. We observed the i -th item at the time instant $t(i) = \frac{i}{n}$. Note that we view items in the order n to 1. We use a thresholding function $p(t)$, such that $p : [0, 1] \rightarrow [0, 1]$ The algorithm works as follows :

When $n \rightarrow \infty$, this translates to viewing the $[nt]^{\text{th}}$ element at time t , for $0 \leq t \leq 1$. We do not pick an item for duration $1/n$ in case it is below the threshold since it translates to picking

Algorithm 3: Optimal Online Decision Procedure

```

1 for  $i = n, \dots, 1$  do
2   if  $F(x_i) \geq p(i/n)$  then
3      $\lfloor$  Select  $x_i$  for the remaining  $\frac{i}{n}$  time ;
4   else
5      $\lfloor$  Do not pick  $x_i$ ;
  
```

127 the item for an infinitesimal duration which does not affect our ASDness guarantee, and hence the
 128 competitive ratio.

129 What remains now is to find the optimal function $p(t)$ that maximises the ASD guarantee.

130

131 10.1 Analysis of the OPT

132 We will calculate the expected time that the optimal holds on to items $\geq x$. This can be obtained
 133 as follows :

$$\mathbb{E}[\text{Time OPT picks } \geq x] = \int_{t=0}^1 \mathbb{P}[\text{OPT has picked } \geq x \text{ at time } t] dt$$

134 Since the OPT picks the maximum seen so far, we can write the above as

$$= \int_{t=0}^1 \mathbb{P}[\max \text{ from } 1 \text{ to } t \text{ is } \geq x] dt$$

135 The max from 1 to t will be distributed as F^{1-t}

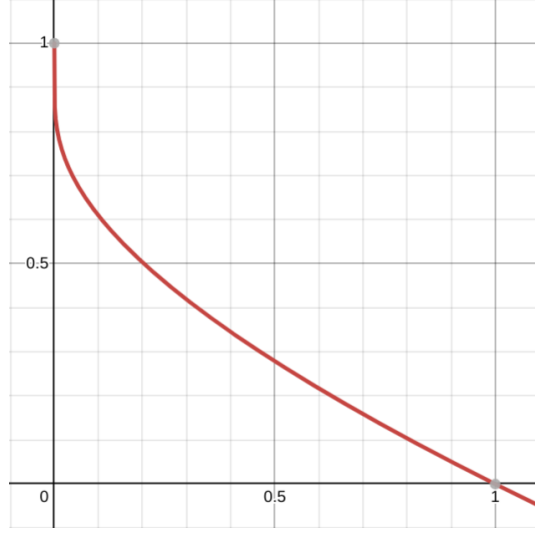
$$\begin{aligned}
 &= \int_{t=0}^1 1 - F^{1-t}(x) dt \\
 &= \int_{t=0}^1 1 - F^t(x) dt \\
 &= 1 - \frac{1 - F(x)}{\ln(\frac{1}{F(x)})}
 \end{aligned}$$

136 Hence we obtain :

$$\mathbb{E}[\text{Time OPT picks } \geq x] = 1 - \frac{1 - F(x)}{\ln(\frac{1}{F(x)})} \tag{11}$$

137 If we plot the curve for this with $F(x)$ on the x-axis, we obtain the following :

138



139

140 The y-axis above represents $\mathbb{E}[\text{Time OPT picks} \geq F^{-1}(x)]$

141 10.2 Analysis of the Algorithm

142 First, it must be the case that the optimal p must be increasing in nature. This is because for
 143 the optimal algorithm we must have that the threshold function matches the optimal dynamic pro-
 144 gramming thresholds obtained in 1, which had increasing thresholds. A similar argument can be
 145 used to claim that $p(0)=0$ for the optimal choice. We also must have $p(1)=1$, this is because **Do this**

146

147 Let us compute the expected time that we hold on to an item $\geq x$. Before that we will define
 148 another quantity of interest, the distribution of the time at which we commit : $q(t)$, where $q(t)$ is
 149 defined as follows : $q(t) = \mathbb{P}[\text{ALG has not committed to an item in the time } (1..t)]$. q and p are
 150 related as follows :

$$\begin{aligned} q(t) - q(t - dt) &= \mathbb{P}[\text{Commit to an item in time } (t - dt \text{ to } t)] \\ &= \mathbb{P}[\text{Not committed to something in } (1, t)] \cdot \mathbb{P}[\text{Commit to } x_t] \\ &= q(t) \cdot (1 - p(t)^{1/n}) \\ &= q(t) \cdot \left(\frac{1}{n} \ln \left(\frac{1}{p(t)} \right) \right) \end{aligned}$$

151 Note that we had chosen $dt = 1/n$ as $n \rightarrow \infty$, and hence the above can be written as :

$$= q(t) \cdot \ln \left(\frac{1}{p(t)} \right) dt$$

152 We obtain the following relation between q and p

$$\frac{q'(t)}{q(t)} = \ln \left(\frac{1}{p(t)} \right) \tag{12}$$

153 Now to calculate $\mathbb{E}[\text{time ALG holds on to } \geq x]$. Let t be the time when the threshold to commit is
 154 x . So $F(x)=p(t)$. If we commit to an item in the time $(1, t)$, then the threshold at that time would
 155 have been higher which guarantees we would have committed to a value greater than x .
 156

$$\begin{aligned}\mathbb{E}[\text{time ALG holds on to } \geq x] &= \sum_{i=\lfloor nt \rfloor}^n \frac{i}{n} \cdot \mathbb{P}[\text{Commit to } i\text{-th item}] \\ &\quad + \sum_{i=0}^{\lfloor nt \rfloor} \frac{i}{n} \cdot \mathbb{P}[\text{Not commit to items } (i \dots n)] \cdot \mathbb{P}[\text{item } i \text{ is } \geq x] \\ &= \sum_{i=\lfloor nt \rfloor}^n \frac{i}{n} \cdot \mathbb{P}[\text{Commit at time } \frac{i}{n}] + \sum_{i=0}^{\lfloor nt \rfloor} \frac{i}{n} \cdot q(\frac{i}{n}) \cdot \mathbb{P}[\text{item } i \text{ is } \geq x]\end{aligned}$$

157 Now we have $\mathbb{P}[\text{Commit at time } \frac{i}{n}] = \frac{1}{n} \cdot q'(\frac{i}{n})$

$$\begin{aligned}&= \sum_{i=\lfloor nt \rfloor}^n \frac{i}{n} \cdot \frac{1}{n} \cdot q'(\frac{i}{n}) + \sum_{i=0}^{\lfloor nt \rfloor} \frac{i}{n} \cdot q(\frac{i}{n}) \cdot \left(1 - F^{\frac{1}{n}}(F^{-1}(p(t)))\right) \\ &= \sum_{i=\lfloor nt \rfloor}^n \frac{i}{n} \cdot \frac{1}{n} \cdot q'(\frac{i}{n}) + \sum_{i=0}^{\lfloor nt \rfloor} \frac{i}{n} \cdot q(\frac{i}{n}) \cdot \left(1 - p(t)^{\frac{1}{n}}\right) \\ &= \sum_{i=\lfloor nt \rfloor}^n \frac{i}{n} \cdot \frac{1}{n} \cdot q'(\frac{i}{n}) + \sum_{i=0}^{\lfloor nt \rfloor} \frac{i}{n} \cdot q(\frac{i}{n}) \cdot \left(\frac{1}{n} \ln \left(\frac{1}{p(t)}\right)\right)\end{aligned}$$

158 We let $n \rightarrow \infty$ and let $\frac{1}{n} = du$, we get :

$$= \int_{u=t}^1 u \cdot q'(u) du + \int_{u=0}^t u \cdot q(u) \cdot \ln \left(\frac{1}{p(t)}\right) du \quad (13)$$

159 Applying integration by parts on the first equation and using $q(1)=1$, we get :

$$\mathbb{E}[\text{time ALG holds on to } \geq x] = 1 - t \cdot q(t) - \int_{u=t}^1 q(u) du + \ln \left(\frac{1}{p(t)}\right) \cdot \int_{u=0}^t u \cdot q(u) du \quad (14)$$

160

161 From 11 we obtained that :

$$\mathbb{E}[\text{Time OPT picks } \geq x] = 1 - \frac{1 - F(x)}{\ln(\frac{1}{F(x)})}$$

162 Since $F(x)=p(t)$, we get that :

$$\mathbb{E}[\text{Time OPT picks } \geq x] = 1 - \frac{1 - p(t)}{\ln(\frac{1}{p(t)})} \quad (15)$$

163 Note that equation 14 is a function of both $p(t)$ and $q(t)$, we instead substitute the following :

$$\begin{aligned}
v(t) &= \ln(q(t)) \\
\implies v'(t) &= \frac{q'(t)}{q(t)} = -\ln(p(t)) \\
\implies p(t) &= -e^{-v'(t)}
\end{aligned} \tag{16}$$

164 Substituting in 14 and 15 we get the following :

$$\mathbb{E}[\text{time ALG holds on to } \geq x] = 1 - te^{v(t)} - \int_{u=t}^1 e^{v(u)} du + v'(t) \int_{u=0}^t ue^{v(u)} du$$

165 and

$$\mathbb{E}[\text{Time OPT picks } \geq x] = 1 - \frac{1 - e^{-v'(t)}}{v'(t)} \tag{17}$$

166 Along with this we have the following boundary conditions on the optimal $v(t)$:

$$\begin{aligned}
v(1) &= \ln(q(1)) = 0 \\
v'(1) &= -\ln(p(1)) = 0
\end{aligned}$$

167 The optimal function $v(t)$ also has the following properties that it will be :

- 168 1. $v(t)$ is an non-decreasing function since $q(t)$ is non-decreasing
- 169 2. $v'(t)$ is a non-increasing function since $p(t)$ is non-decreasing

170 Now say the optimal $p(t)$ guarantees a k -ASD guarantee. We **claim** that this would imply that
171 $\forall 0 \leq x :$

$$\mathbb{E}[\text{time ALG holds on to } \geq x] = k \cdot \mathbb{E}[\text{Time OPT picks } \geq x]$$

172 ie. $\nexists 0 \leq x \leq 1$ such that

$$\mathbb{E}[\text{time ALG holds on to } \geq x] > k \cdot \mathbb{E}[\text{Time OPT picks } \geq x]$$

173 If there were such an

174 **Show that Abel Et al.'s algorithm is not optimal**

175 **A Expectation Formula**

176 The expected value of a random variable with distribution F can be expressed as follows :

$$\mathbb{E}[X] = \int_{x=0}^{\infty} (1 - F(x))dx - \int_{x=-\infty}^0 F(x)dx \quad (18)$$

177 This can be proved as follows :

$$\mathbb{E}[X] = \int_{x=-\infty}^0 xf(x)dx + \int_{x=0}^{\infty} xf(x)dx$$

178 Computing each term separately and applying integration by-parts, we get:

$$\begin{aligned} \int_{x=-\infty}^0 xf(x)dx &= [xF(x)]_{-\infty}^0 - \int_{x=-\infty}^0 F(x)dx \\ &= - \int_{x=-\infty}^0 F(x)dx \end{aligned} \quad (19)$$

179 and

$$\begin{aligned} \int_{x=0}^{\infty} xf(x)dx &= [x(F(x) - 1)]_0^{\infty} - \int_{x=0}^{\infty} (F(x) - 1)dx \\ &= \int_{x=0}^{\infty} (1 - F(x))dx \end{aligned} \quad (20)$$

180 Adding eq. (19) and eq. (20), we get eq. (18)

(21)

181 Moreover, if the random variable X has a positive support ie. $F(x) = 0 \forall x < 0$, then eq. (18) can
182 be simplified to

183

$$\mathbb{E}[X] = \int_{x=0}^{\infty} (1 - F(x))dx \quad (22)$$

184 B Proof of optimality

185 C ASDness equivalence to Competitiveness

186 C.1 A c-ASD algorithm is also c-Competitive

187 A c-ASD algorithm guarantees the following :

$$\mathbb{E}[\text{time ALG chooses} \geq x] \geq c \cdot \mathbb{E}[\text{time OPT chooses} \geq x]$$

188 Let $G(x) = \mathbb{E}[\text{time ALG chooses} \geq x]$, then $1 - G(x) = \mathbb{E}[\text{time ALG chooses} < x]$. If we assume
 189 that there are no points masses in our distribution, then $1 - G(x) = \mathbb{E}[\text{time ALG chooses} < x]$, this
 190 in turn represents the CDF of a random variable that denotes the time-distribution for which we
 191 hold on to an item with value x . The expected value of this random variable is the weighted sum
 192 of items value and the time for which we hold on to the item, which represents the pay off of the
 193 algorithm. From eq. (22) we know that this can be represented as :

$$\begin{aligned} \int_{x=0}^{\infty} (1 - (1 - G(x)))dx &= \int_{x=0}^{\infty} G(x)dx \\ \implies \mathbb{E}[ALG] &= \int_{x=0}^{\infty} \mathbb{E}[\text{time ALG chooses} \geq x]dx \\ &\geq c \cdot \int_{x=0}^{\infty} \mathbb{E}[\text{time OPT chooses} \geq x]dx \end{aligned}$$

194 With a similar argument as above we can also show that

$$\mathbb{E}[OPT] = \int_{x=0}^{\infty} \mathbb{E}[\text{time OPT chooses} \geq x]dx$$

195 And therefore a c-ASD algorithm guarantees that

$$\mathbb{E}[ALG] \geq c \cdot \mathbb{E}[OPT] \tag{23}$$

196 and hence it is c-Competitive.

197 C.2 A c-competitive algorithm allows for a c-ASD algorithm

198 We will prove this for distributions that have a finite support. We will prove that the contrapositive
 199 holds true, ie. if there does not exist an algorithm that is c-ASD, then there does not exist an
 200 algorithm that is c-Competitive. Let the support of the distribution F be the finite set $a_1, \dots, a_m$.
 201 Any deterministic algorithm merely sets a threshold $\tau(i)$ for each item i , these thresholds could
 202 depend on the past realisations seen by the algorithm, however note that all thresholds $\tau(i)$ lying in
 203 the range $[0, a_1]$ or $(a_{j-1}, a_j]$ behave identically to any other threshold within the same range. Thus
 204 there are only a finite set of distinct deterministic Algorithms call them $A_1, A_2, \dots, A_N$ possible. A

205 general randomised algorithm can be seen as behaving as A_i with probability $\lambda_i \geq 0$ for $i = 1, \dots, N$.
 206 Suppose no Algorithm is c-ASD then we have the following guarantee :

$$\begin{aligned} & \max_{\lambda_1, \dots, \lambda_N} \min_{a_1, a_2, \dots, a_m} \frac{\mathbb{E}[\text{days ALG picks} \geq a_j]}{\mathbb{E}[\text{days OPT picks} \geq a_j]} < c \\ \implies & \max_{\lambda_1, \dots, \lambda_N} \min_{a_1, a_2, \dots, a_m} \sum_{i=1}^N \frac{\mathbb{E}[\text{days } A_i \text{ picks} \geq a_j]}{\mathbb{E}[\text{days OPT picks} \geq a_j]} < c \end{aligned}$$

207 This can be written in the form of an LP as follows :

$$\begin{aligned} & \text{maximise } \alpha, \quad \text{subject to :} \\ & \mathbb{E}[\text{days OPT picks} \geq a_j] \cdot \alpha - \sum_{i=1}^N \lambda_i \cdot \mathbb{E}[\text{days } A_i \text{ picks} \geq a_j] \leq 0 \quad \forall j = 1, \dots, m \\ & \sum_{i=1}^N \lambda_i = 1 \\ & \lambda_i \geq 0 \quad \forall i = 1, \dots, N \end{aligned}$$

208 The Dual of the above LP is the following :

$$\begin{aligned} & \text{minimise } z, \quad \text{subject to :} \\ & \sum_{j=1}^m y_j \cdot \mathbb{E}[\text{days OPT picks} \geq a_j] = 1 \\ & z - \sum_{j=1}^m y_j \cdot \mathbb{E}[\text{days } A_i \text{ picks} \geq a_j] \geq 0 \quad \forall i = 1, 2, \dots, N \\ & y_j \geq 0 \quad \forall j = 1, 2, \dots, m \end{aligned}$$

209 say z attains the minimum z^* at $y_1^*, y_2^*, \dots, y_m^*$, then we have that :

210 References

211 [1] Andreas Abels, Elias Pitschmann, and Daniel Schmand. Prophet-inequalities over time, 2022.